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Supplementary Materials for

Topic choice contributes to the lower rate of NIH awards to African-American/black scientists

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The PDF file includes:

Table S1. Attributes of R01 submissions by race of applicant.
Table S2. Distribution of AA/B and WH PIs across institutional funding quintiles.
Table S3. Comparison of the proportion of applications from new investigators by race (R01 Types 1 and 2, FY 2011–2015).
Table S4. Comparison of impact scores for R01 applications from AA/B and WH applicants (Types 1 and 2, FY 2011–2015).
Table S5. Comparison of word2vec topic assignment to human SME annotation.
Table S6. Statistically significant variation in the award rates of topic clusters.
Table S7. Topics favored by AA/B applicants compared to topics with no AA/B applicants.
Table S8. Effect of topic choice on funding gap throughout the R01 application process.
Table S9. Effect of removing variables from regression models (awarded given discussed).
Table S10. Reviewer demographics for all study section meetings that considered R01 applications (FY 2011–2015).
Fig. S1. Racial differences in the number of unique R01 applications submitted per applicant (Types 1 and 2, FY 2011–2015).
Fig. S2. Funding gap between AA/B and WH scientists at each stage of the R01 application and review process at institutions in the highest and lowest NIH funding quintiles.
Fig. S3. Comparison of career age distributions for AA/B and WH scientists.
Fig. S4. Time between initial application submission and resubmission.
Fig. S5. Distribution of topics across study sections.
Fig. S6. Comparison of topic choice variation by race.
Fig. S7. Award rates by topic cluster size.
Fig. S8. Scientific influence for higher and lower success topics.
Fig. S9. Comparison of scientific influence for publications linked to awards from higher and lower success topics by percentile score.
Fig. S10. Scientific influence of publications by new investigators before and after receiving their first award.

Other Supplementary Material for this manuscript includes the following:

(available at advances.sciencemag.org/cgi/content/full/5/10/eaaw7238/DC1)

Regression analysis results for all variables (Microsoft Excel format)

SUPPLEMENTAL TABLES

Table S1. Attributes of R01 submissions by race of applicant. All Type 1 and 2 R01s submitted between FY2011-FY2015 were considered in this analysis (overall N=157,549; N=106,023 for applications from AA/B and WH applicants). N is the number of applications in each category. For each category, the number of applications is given in parentheses following the percent represented (e.g., 10.7%, or 256 applications out of 2403 were awarded for AA/B applicants).Data for unknown, withheld, or racial/ethnic categories other than AA/B or WH are excluded from this table.

	N applications	AA/B N=2403	White N=103,620
Awarded	18,571	10.7% (256)	17.7% (18,315)
Discussed	60,523	44.0% (1057)	57.4% (59,466)
Percentile score if discussed			
Lower quartile bound		17	13
Median		33	26
Upper quartile bound		45	40
Animal and Human study	10,343	7.6% (183)	9.8% (10,160)
Animal only study	43,987	32.1% (772)	41.7% (43,215)
Human only study	34,108	49.8% (1197)	31.8% (32,911)
Neither Animal nor Human study	17,585	10.4% (251)	16.7% (17,334)
Degree: PhD	64,770	54.9% (1320)	61.2% (63,450)
Degree: MD	13,740	15.8% (379)	12.9% (13,361)
Degree: MD-PhD	8,545	8.1% (194)	8.1% (8351)
Degree: Other doctoral	1783	3.4% (81)	1.6% (1702)
New investigator	26,934	47.4% (1138)	24.9% (25,796)
Multi-PI application	20,358	19.3% (464)	19.2% (19,894)

Table S2. Distribution of AA/B and WH PIs across institutional funding quintiles.

Institutional funding quintile	AA/B PIs (N=948)	% AA/B PIs	WH PIs (N=34,218)	% WH PIs
1 (highest)	185	19.5%	6652	19.4%
2	133	14.0%	6422	18.8%
3	138	14.6%	6758	19.7%
4	171	18.0%	6864	20.1%
5 (lowest)	321	33.9%	7522	22.0%

Table S3. Comparison of the proportion of applications from new investigators by race (R01 Types 1 and 2, FY 2011–2015).

	AA/B	WH
Submitted applications	2403	103,620
Applications from new investigators	1138	25,796
% of applications from new investigators	47.4%	24.9%

1.9-fold difference

Table S4. Comparison of impact scores for R01 applications from AA/B and WH applicants (Types 1 and 2, FY 2011–2015).

Applicant race	Total applications	Scored applications	Percent scored	Average score	Std. Dev.	Minimum score	Maximum score
AA/B	2403	1056	44%	38.4	13.4	10	90
WH	103,620	59,453	57%	35.2	12.6	10	90

Table S5. Comparison of word2vec topic assignment to human SME annotation. Results are presented in order of topics determined by word2vec to be least similar (similarity 0.0) to most similar (similarity 0.9). The maximum number of applications that could be partitioned to each topic was 25 (5 SMEs, 5 applications per topic A or B in each similarity group).

word2vec content similarity	Cluster numbers	word2vec description	Human annotated as A	Human annotated as B
0.0	A - 26	physical_activity overweight weight lifestyle recommendation weight_loss BMI nutrition weight_gain education	25	0
	B - 145	mRNA RNA transcript post-transcriptional miRNA microRNA gene_expressionbinding_proteins non-coding protein	0	25
0.1	A - 30	racial minority MeSH_African_Americanhispanicamerican woman disparity African-American ethnic socioeconomic	25	0
	B - 56	retina photoreceptor retinal cone MeSH_Photoreceptor_Cells_Vertebrate rod photoreceptor_cellsretinal_degenerationMeSH_Retinal_NeuronMeSH_Retinal_Cone_Photoreceptor_Cell	0	25
0.2	A - 35	bone_mineral_densitybone_density fracture BMD bone osteoporosi DXA trabecular_bone microarchitecture bone_loss	25	0
	B - 93	MeSH_T-Lymphocyte CD8 anti-tumor antigen-specific t antigen tumor-specific immune_response immunotherapy	0	25
0.3	A - 0	GWA variant MeSH_Genetic_Association_Studygenetic_variants genotyping genome-wide MeSH_Genome-Wide_Association_Study locus case-control cohort	25	0
	B - 68	microbial periodontal bacterium bacterial MeSH_Dental_Plaque periodontitis periodontal_disease biofilm MeSH_Porphyromonas_gingivaligingivali	0	25
0.4	A - 72	parasite malarium plasmodium MeSH_Plasmodium_falciparum falciparum parasitic p. malarial MeSH_Parasitic_Diseasevivax	24	1
	B - 142	mitotic mitosicell_division chromosome MeSH_Spindle_ApparatuMeSH_Chromosome_Segregationcell_cycle spindle centrosome cell_cycle_progression	1	24
0.5	A - 55	axon axonal spinal_cord neuron neuronal nerve neuriteMeSH_Cell_Body outgrowth synapse	25	0
	B - 47	circadian circadian_clockMeSH_Circadian_Rhythm clock rhythm circadian_rhythm rhythmicity circadian_clocksMeSH_Circadian_Clockcircadian_rhythms	0	25
0.6	A - 58	pet image tracer radiotracer SPECT biodistributionpositron_emission_tomography radiopharmaceutical molecular_imaging radiolabeled	25	0
	B - 131	compound small_moleculeMeSH_Structure-Activity_Relationshipsmall_molecules inhibitor potency library MeSH_Chemistry_Pharmaceutical drug-like MeSH_Small_Molecule_Library	0	25
0.7	A - 3	software MeSH_Software researcher MeSH_Statistics_as_Topic database informatic dataset databasis algorithm analytical	24	1
	B - 4	analytical detection platform technology multiplexed instrumentation microfluidic capture device high_throughput	1	24
0.8	A - 103	inflammatory proinflammatory pro-inflammatory innate_immune_response leukocyte inflammatory_mediatorsMeSH_Toll-Like_Receptor inflammation cytokine	25	0
	B - 104	airway_inflammation asthmatic allergic airway allergen th2 hyperresponsivenes asthma atopic	0	25
0.9	A - 80	MeSH_Breast_Neoplasmbreast_cancerstumor_cellsbreast_cancer_cellstumor_progression mammary breast_cancercancer_cells tumor mammary_tumors	21	4
	B - 85	oncogenic oncogene tumor_cells PI3K cancer_cell AKT transform tumor_progression MEK pathway	4	21

Table S6. Statistically significant variation in the award rates of topic clusters. 56 topic clusters have award rates that differ significantly ($p<0.01$) from the average award rate of 16.3%. Clusters are ranked by award rate; 25 are above and 31 are below the average. Clusters highlighted in blue and yellow appear in the word clouds in Figure 4A and B, respectively. Some word2vec descriptions have been truncated due to space constraints.

cluster number	award rate	p-value (<x)	word2vec description
Clusters above the average award rate			
42	28.7%	10^{-11}	odor olfactory olfactory_bulb olfactory_system odorant chemosensory sensory circuit neuron gustatory
56	27.7%	10^{-6}	retina photoreceptor retinal cone MeSH_Photoreceptor_Cells_Vertebrate rod photoreceptor_cells retinal_degeneration
108	26.6%	10^{-5}	cornea corneal len eye ocular corneal_epithelium ocular_surface lens corneal_stroma conjunctiva
54	26.0%	10^{-5}	glaucoma IOP MeSH_Intraocular_Pressure glaucomatous optic_nerve retinal retinal_ganglion_cells optic_nerve_head eye RGC
9	25.8%	10^{-15}	perceptual representation visual visual_stimuli task neural stimulus object cortical psychophysical
143	24.4%	10^{-14}	RNA mRNA protein ribosome machinery binding_proteins tRNA eukaryotic transcript transcription
10	24.2%	10^{-5}	speech auditory sound perceptual acoustic speech_perception representation psychophysical hear perception
101	23.6%	0.01	iron in_iron heme fe ferritin transferrin hepcidin ferritin-bound transport transferrin-mediated
142	23.0%	10^{-6}	mitotic mitosis cell_division chromosome MeSH_Spindle_Apparatu MeSH_Chromosome_Segregation cell_cycle spindle
128	22.9%	10^{-15}	conformation bound conformational crystallographic structure structural_features MeSH_Crystallography_X-Ray
57	22.9%	0.001	cochlea cochlear hair_cells hair_cell MeSH_Organ_of_Cortus inner_ear ear vestibular hear OHC
129	22.6%	10^{-5}	catalytic active_site reaction electron_transfer catalysis enzyme MeSH_Catalytic_Domain enzymatic spectroscopic substrate
137	22.4%	10^{-15}	cytoskeletal cytoskeleton actin MeSH_Cell_Movement MeSH_Actin_Cytoskeleton MeSH_Cell_Adhesion adhesion motility
43	22.0%	10^{-14}	synaptic synapse neuron neuronal synaptic_plasticity activity-dependent presynaptic excitatory MeSH_Synaptic_Transmission
55	21.7%	10^{-4}	axon axonal spinal_cord neuron neuronal nerve neurite MeSH_Cell_Body outgrowth synapse
140	21.5%	10^{-5}	MeSH_DNA_Repair DSB homologous_recombination DNA MeSH_DNA_Damage chromosome MeSH_DNA_Break
139	21.3%	0.01	cilium ciliary primary_cilia ciliogenesis polycystin ciliagenesis polycystic_kidney_disease morphogenesis primary_cilium
141	21.2%	0.001	DNA polymerase MeSH_DNA_Repair mutagenesis MeSH_DNA-Directed_DNA_Polymerase nucleotide strand
127	21.1%	0.001	catalyst synthesis synthetic heterocycle chiral chemistry MeSH_Chemistry_Techniques_Synthetic enantioselective
138	20.9%	10^{-6}	embryonic patterning morphogenesis embryogenesis embryo neural_crest fate specification morphogenetic
135	20.7%	10^{-5}	progenitor progenitor_cells differentiation lineage fate embryonic stem_cells stem_cell hematopoietic progenitor_cell
144	20.4%	10^{-6}	chromatin transcription transcriptional histone chromatin_modifications MeSH_Transcriptional_Activation
149	20.4%	10^{-4}	ubiquitin ubiquitination ubiquitin_ligase degradation MeSH_Ubiquitin-Protein_Ligase proteasomal protein proteasome
76	20.4%	0.01	gene_therapy vector gene_transfer MeSH_Dependovirus transgene recombinant AAV lentiviral_vectors rAAV transduced
132	18.4%	0.01	genome genomic MeSH_Base_Sequence variation locus whole_genome whole-genome genome-wide sequence catalog

cluster number	award rate	p-value (<x)	word2vec description
Clusters below the average award rate			
27	14.5%	0.01	socioeconomic health_outcomes demographic policy economic MeSH_Data_Collection individual-level well-being household
32	14.0%	10^{-5}	practice provider clinician care education evidence-based health_care recommendation medical psychosocial
28	14.0%	0.001	youth child adolescent psychosocial substance_use mental_health educational social adolescence MeSH_Risk-Taking
131	13.9%	0.01	compound small_molecule MeSH_Structure-Activity_Relationship small_molecules inhibitor potency library
118	13.7%	0.01	insulin_sensitivity adipose_tissue insulin_resistance fat glucose triglyceride lipid_metabolism fatty_acid diet-induced
44	13.7%	0.01	AD amyloid_deposition alzheimer amyloid_beta neurodegeneration beta-amyloid amyloid-betum MeSH_Plaque_Amyloid
74	13.6%	0.001	vaccine epitope MeSH_Antibody_Formation immunogenicity antigen immunogen immunization neutralize vaccination
121	13.4%	0.01	cardiac myocardium myocardial heart myocyte cardiomyocyte myocardial_infarction remodel cardiac_myocytes
6	13.4%	0.01	image accuracy scanner MRus treatment_planning phantom accurate radiation CT image_quality
83	13.4%	0.01	glioma glioblastoma tumor GBM tumor_cells MeSH_Brain_Neoplasm MeSH_Glioblastoma malignant_gliomas brain_tumors
94	13.2%	0.01	autoantibody autoimmune autoimmunity MeSH_B-Lymphocyte MeSH_T-Lymphocyte autoreactive cytokine
36	13.1%	10^{-5}	cohort risk MeSH_Prospective_Study MeSH_Cohort_Study risk_factors prospective incident MeSH_Epidemiologic_Study
4	12.8%	10^{-4}	analytical detection platform technology multiplexed instrumentation microfluidic capture device high_throughput
134	12.5%	0.01	MeSH_DNA_Methylation methylation genome-wide epigenetic gene_expression methylome genome_wide
21	12.4%	0.01	pollutant exposure pollution MeSH_Environmental_Exposure air indoor contaminant air_pollution MeSH_Epidemiologic_Study
26	12.2%	10^{-5}	physical_activity overweight weight lifestyle recommendation weight_loss BMI nutrition weight_gain education
62	11.9%	10^{-6}	biomaterial tissue_engineering material hydrogel implant biomimetic fabrication tissue_engineered porous
136	11.5%	10^{-5}	pluripotent pluripotent_stem_cells embryonic_stem_cells embryonic_stem MeSH_Induced_Pluripotent_Stem_Cell
30	11.2%	10^{-9}	racial minority MeSH_African_American hispanic american woman disparity African-American ethnic socioeconomic
45	10.8%	10^{-4}	dopaminergic dopaminergic_neurons neurodegeneration nigrostriatal substantia_nigra striatal dopamine striatum neuron
33	10.8%	10^{-4}	pregnancy maternal gestation fetal placental fetus placenta intrauterine pregnant birth
59	10.7%	10^{-11}	nanoparticle conjugate payload delivery nanocarrier biodistribution liposome polymeric polymer liposomal
34	10.6%	10^{-8}	maternal prenatal offspring birth pregnancy in_utero perinatal gestation mother exposure
75	10.6%	10^{-4}	tuberculosis MeSH_Mycobacterium_tuberculosis TB mycobacterial mycobacterium bacilli MTB bacterial
37	10.4%	10^{-6}	exercise physical_function decline muscle_strength muscle_mass quality_of_life older_adults endurance older
82	10.2%	10^{-17}	tumor_cells tumor primary_tumors metastatic malignant metastasi primary_tumor tumor_progression metastase cancer_cells
81	9.8%	10^{-7}	prostate_cancer prostate PCa prostate_cancer_cells prostate_cancers castration-resistant CRPC prostatic androgen_receptor
80	9.7%	10^{-15}	MeSH_Breast_Neoplasm breast_cancers tumor_cells breast_cancer_cells tumor_progression mammary breast_cancer
63	9.7%	10^{-4}	HPV MeSH_Human_papillomavirus_16 cervical cervical_cancer HPV-associated human_papilloma_virus papillomavirus
119	9.0%	10^{-4}	prostaglandin arachidonic_acid fatty_acids PUFA eicosanoid COX-2 MeSH_Dinoprostone fish_oil cyclooxygenase omega-3
97	7.5%	10^{-5}	ovary reproductive ovulation granulosa_cells follicle oocyte ovarian gonadotropin MeSH_Ovarian_Follicle fertility

Table S7. Topics favored by AA/B applicants compared to topics with no AA/B applicants. Topic clusters are ranked from lowest to highest overall award rate.

Cluster number	word2vec description	N apps	% apps AA/B	Overall award rate	Award rate quintile	% apps funded AA/B	% apps funded WH	Relative funding disparity WH:AA/B
Clusters with highest percentages of applications from AA/B applicants (overall funding rates between 11.2% and 17.2%)								
30	racial minority MeSH_African_American hispanic american woman disparity African-American ethnic socioeconomic	1908	8.8%	11.2%	5	13.2%	12.7%	0.96
26	physical_activity overweight weight lifestyle recommendation weight_loss BMI nutrition weight_gain education	1659	3.7%	12.2%	5	4.8%	13.5%	2.79
36	cohort risk MeSH_Prospective_Study MeSH_Cohort_Study risk_factors prospective incident MeSH_Epidemiologic_Study incidence baseline	2683	2.7%	13.1%	5	5.5%	14.1%	2.57
32	practice provider clinician care education evidence-based health_care recommendation medical psychosocial	5125	2.8%	14.0%	4	13.1%	14.9%	1.14
28	youth child adolescent psychosocial substance_use mental_health educational social adolescence MeSH_Risk-Taking	3269	3.1%	14.0%	4	11.9%	14.7%	1.24
27	socioeconomic health_outcomes demographic policy economic MeSH_Data_Collection individual-level well-being household health	2376	4.2%	14.5%	4	6.0%	16.6%	2.76
31	health_care care practice provider healthcare medical physician clinician service hospital	3270	2.3%	15.3%	3	9.2%	16.3%	1.77
29	care service MeSH_South_Africa country HIV provider woman practice youth MeSH_Risk-Taking	2469	7.2%	17.2%	2	12.4%	19.4%	1.56
Clusters with no applications from AA/B applicants (overall funding rates between 12.5% and 28.7%)								
60	cartilage articular_cartilage knee joint OA osteoarthritis articular MeSH_Knee_Joint tissue_engineered_cartilage biomechanical	457	0.0%	12.5%	5	-	9.8%	-
87	skin keratinocyte epidermal epidermis cutaneous UVB normal_skin melanocyte tetrachlorodibenzodioxin dermal	446	0.0%	14.6%	4	-	15.5%	-
115	aortic valve valvular wall aortic_valve calcification valf vessel artery aortum	346	0.0%	15.0%	4	-	15.9%	-
84	leukemic hematopoietic leukemia AML leukemic_cells myeloid acute_myeloid_leukemia hematopoietic_cells hematopoietic_stem_cells	898	0.0%	16.0%	3	-	18.6%	-
126	prion PrPSc PrP MeSH_Prion_Disease prion_protein self-templating PrP23-144 aggregate MeSH_Amyloid transmissible_spongiform_encephalopathies	218	0.0%	18.8%	2	-	18.8%	-
101	iron in_iron heme fe ferritin transferrin hepcidin ferritin-bound transport transferrin-mediated	203	0.0%	23.6%	1	-	26.8%	-
108	cornea corneal len eye ocular corneal_epithelium ocular_surface lenses corneal_stroma conjunctiva	267	0.0%	26.6%	1	-	32.2%	-
42	odor olfactory olfactory_bulb olfactory_system odorant chemosensory sensory circuit neuron gustatory	495	0.0%	28.7%	1	-	28.5%	-

Table S8. Effect of topic choice on funding gap throughout the R01 application process.

	Average marginal effects (AMEs) for race[1]		Mean rate for WH applicants[3]	% of mean rate for WH applicants		% decrease in gap when topic included[4]	p-value[5]
Application stage	Topic not inlcuded	Topic included[2]		Topic not included	Topic included		
Awarded overall	-0.0357***	-0.0329***	0.149	-24.04	-22.15	7.8	0.010
	(0.00723)	(0.00732)					
Discussed	-0.0823***	-0.0860***	0.537	-15.31	-16.00	-4.5	0.025
	(0.0137)	(0.0137)					
Awarded given discussed	-0.0322*	-0.0254	0.276	-11.66	-9.19	21.1	0.005
	(0.0149)	(0.0151)					

[1]The AMEs are the percentage point differences for award rates for AA/B applicants as compared to the base award rate for WH applicants. Standard errors for AMEs are reported in parentheses. AMEs are statistically significant if indicated with asterisks (*p<0.05, ***p<0.001).

[2]Topic areas were parameterized into 89 superclusters to generate the values in this table (see Materials and Methods).

[3]The mean awarded or discussed rate for WH applicants, used as baseline

[4]Percent decrease in the funding gap between AA/B and WH applicants when including topic area

[5]p-values relatingto whether the coefficients of the underlying regression models with and without topic are significantly different

Table S9. Effect of removing variables from regression models (awarded given discussed).

Removed variable(s)	AME for race[1]	% gap decrease when removing variable(s)[2]	p-value[3]
None (base model with all variables including topic	-0.0254	---	
Fiscal Year	-0.0261	2.6	0.216
Median RCR (IHS transformed)	-0.0273	6.9	0.038
Top-decile RCR publication count (IHS transformed)	-0.0259	1.8	0.150
Both RCR variables	-0.0344	26.1	<0.001
Application is a resubmission	-0.0295	14.1	0.473
Years since applicant's last degree	-0.0271	6.5	0.024
Applicant is an ESI	-0.0242	-5.1	0.240
Both career stage variables	-0.0226	-12.5	0.038
Organization region (categorical)	-0.0260	2.3	0.671
Organization Carnegie classification (categorical)	-0.0253	-0.4	0.717
Organization type (IMPAC II; categorical)	-0.0252	-0.6	0.580
All organization categorical variables	-0.0260	2.3	0.680
Organization R01 funding (IHS-transformed;	-0.0251	-1.2	0.360
Organization PI counts (IHS-transformed; continuous)	-0.0248	-2.2	0.153
Both organization continuous variables	-0.0252	-0.7	0.729
All organization variables	-0.0262	3.2	0.585
Number of prior R01 applications	-0.0317	19.8	0.007
Number of prior R01 awards	-0.0292	12.9	0.015
Both prior applications and awards	-0.0319	20.3	0.005

[1]The AMEs are the percentage point differences for award rates for AA/B applicants as compared to the base award rate for WH applicants.

[2]Percentage the funding gap decreases when comparing the base model without relevant variables to the base model with all the variables. The difference in AMEs between the model without the indicated variable(s) and the base model was determined, and this value was divided by the AME of the model without the indicated variable(s) to determine the percentage decrease in the funding gap.

[3]p-values relating to whether the coefficients of the underlying regression models with and without the added variables are significantly different

Table S10. Reviewer demographics for all study section meetings that considered R01 applications (FY 2011–2015). Each meeting attendance by a reviewer is counted as a separate event. Percentages are based on reviewers with reported demographic data; those with missing or unknown race were excluded.

Reviewer race	% reviewers (N=103,827)
AA/B	2.4
WH	77.8
Other	19.8

SUPPLEMENTAL FIGURES

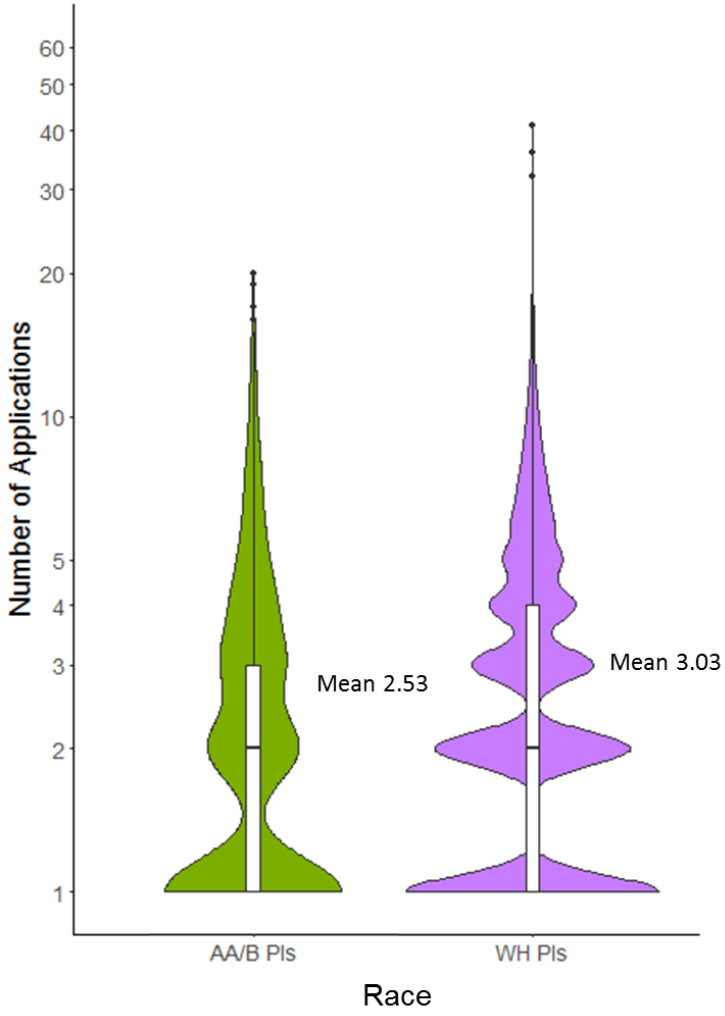
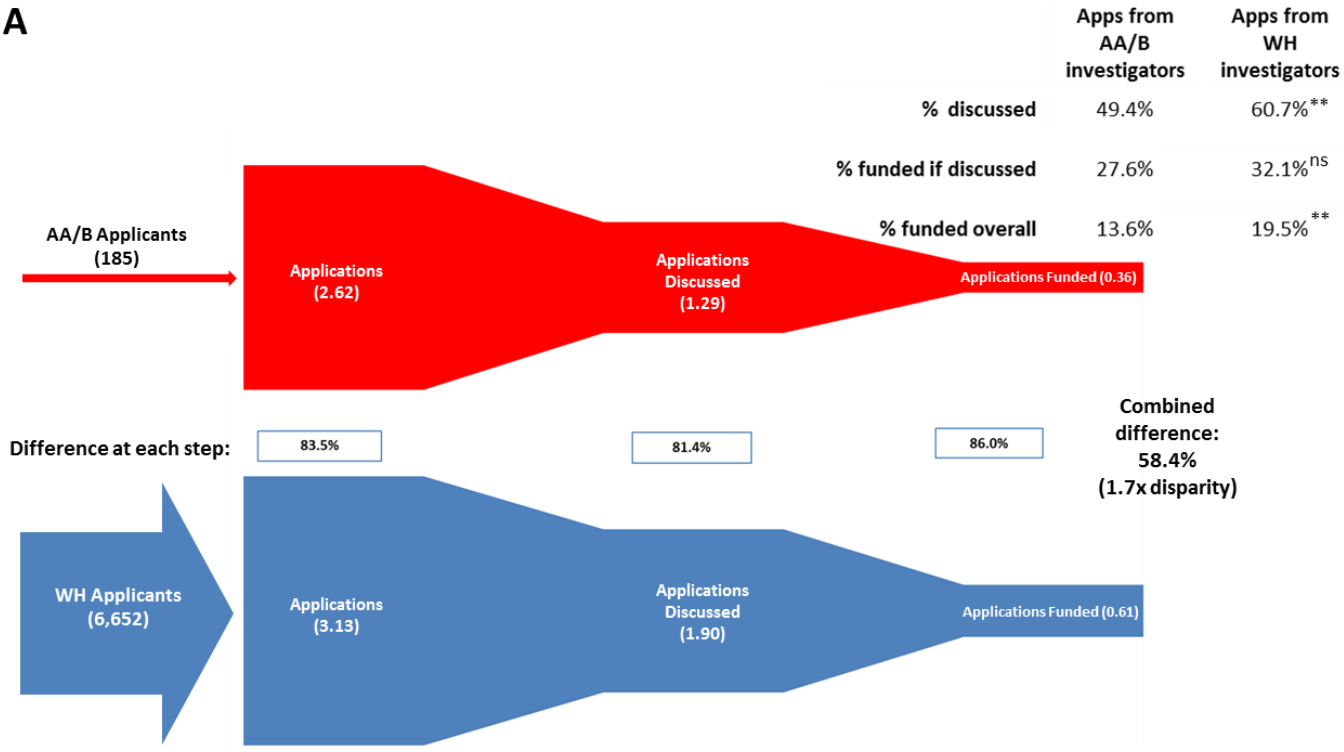


Fig. S1. Racial differences in the number of unique R01 applications submitted per applicant (Types 1 and 2, FY 2011–2015).

A


Apps from AA/B investigators
Apps from WH investigators
% discussed
49.4%
60.7%**
% funded if discussed
27.6%
32.1%ns
% funded overall
13.6%
19.5%**
AA/B Applicants (185)
Applications (2.62)
Applications Discussed (1.29)
Applications Funded (0.36)
Difference at each step:
83.5%
81.4%
86.0%
Combined difference: 58.4% (1.7x disparity)
WH Applicants (6,652)
Applications (3.13)
Applications Discussed (1.90)
Applications Funded (0.61)

B

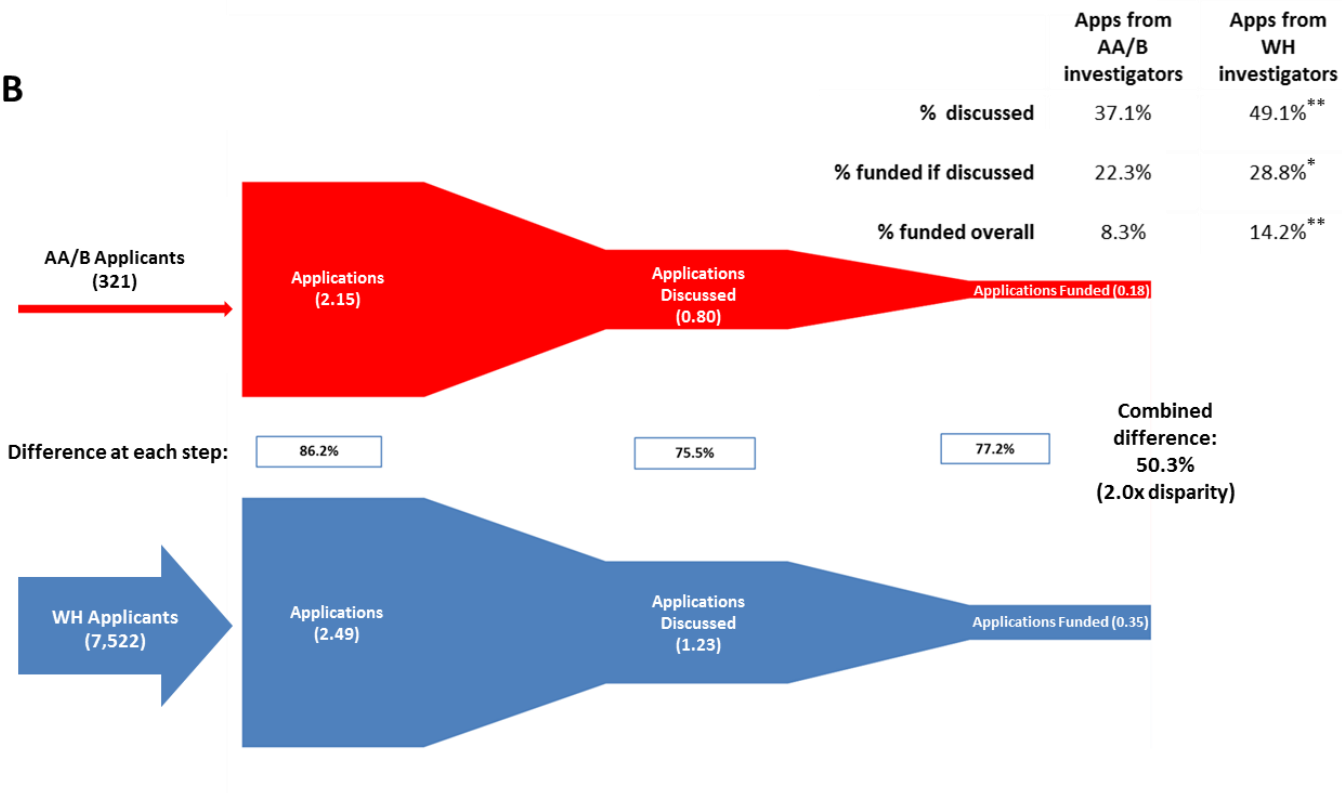

Apps from AA/B investigators
Apps from WH investigators
% discussed
37.1%
49.1%**
% funded if discussed
22.3%
28.8%*
% funded overall
8.3%
14.2%**
AA/B Applicants (321)
Applications (2.15)
Applications Discussed (0.80)
Applications Funded (0.18)
Difference at each step:
86.2%
75.5%
77.2%
Combined difference: 50.3% (2.0x disparity)
WH Applicants (7,522)
Applications (2.49)
Applications Discussed (1.23)
Applications Funded (0.35)

Fig. S2. Funding gap between AA/B and WH scientists at each stage of the R01 application and review process at institutions in the highest and lowest NIH funding quintiles. (**A**) Highest funding quintile, (**B**) Lowest funding quintile. The number of AA/B (red) and WH (blue) applicants is presented in the arrows on the left. Number of applications per applicant that were submitted, discussed, and funded are presented in the rocket charts. Comparative rates of discussion, funding of discussed applications, and overall funding rates are presented in the summary table on the top right (**p<0.01, *p<0.05, ns = not statistically significant).

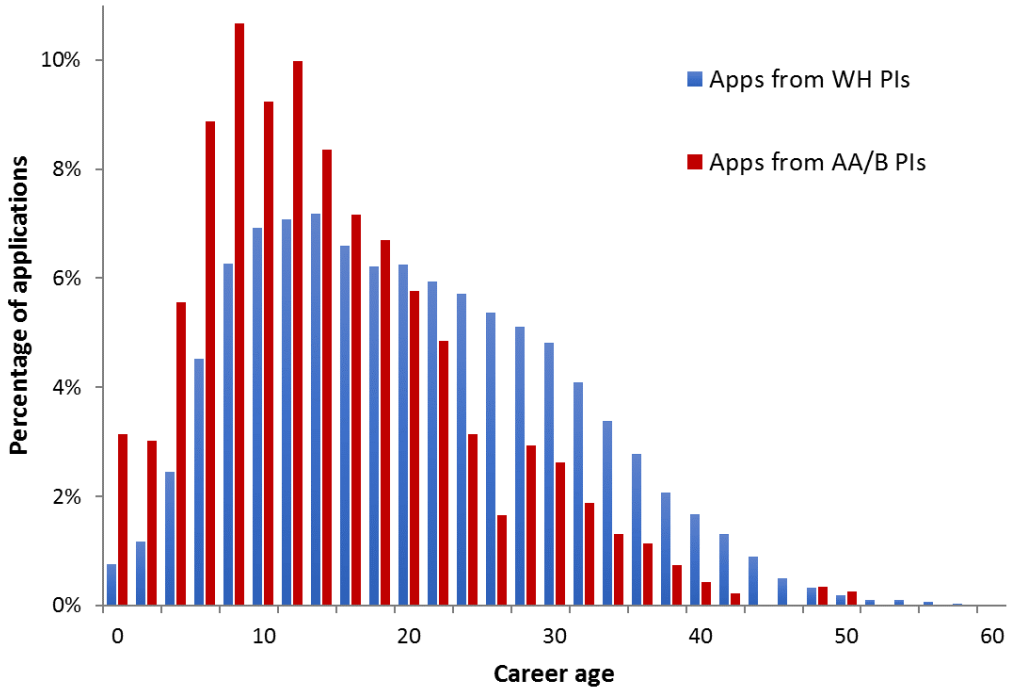


Fig. S3. Comparison of career age distributions for AA/B and WH scientists. Career age for the PI of each application (FY2011-FY2015) was determined by subtracting PI terminal degree year from the application fiscal year.

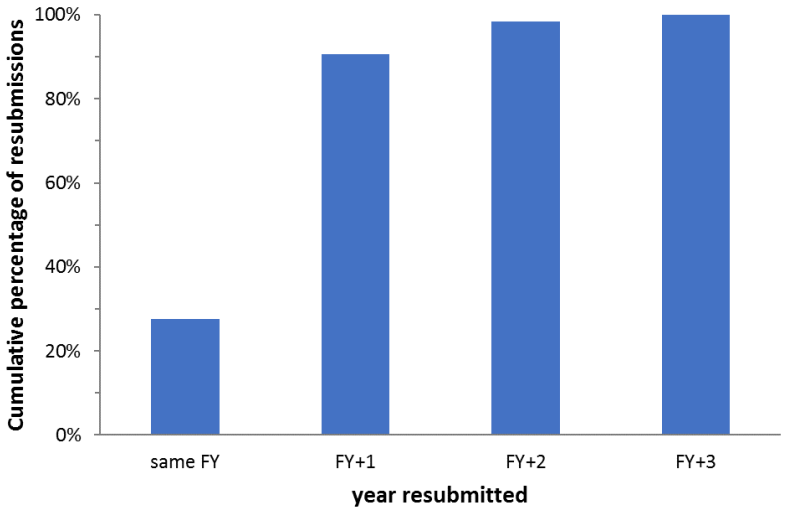


Fig. S4. Time between initial application submission and resubmission. Time between initial submission (for R01 applications from FY2011-2014) and resubmission (FY2011-2017) was assessed for a three-year window after initial submission, since resubmissions must be received within 37 months of initial application submission (*10, 11*). FY2015 applications were excluded from this analysis, as only two years had elapsed for their initial submissions.

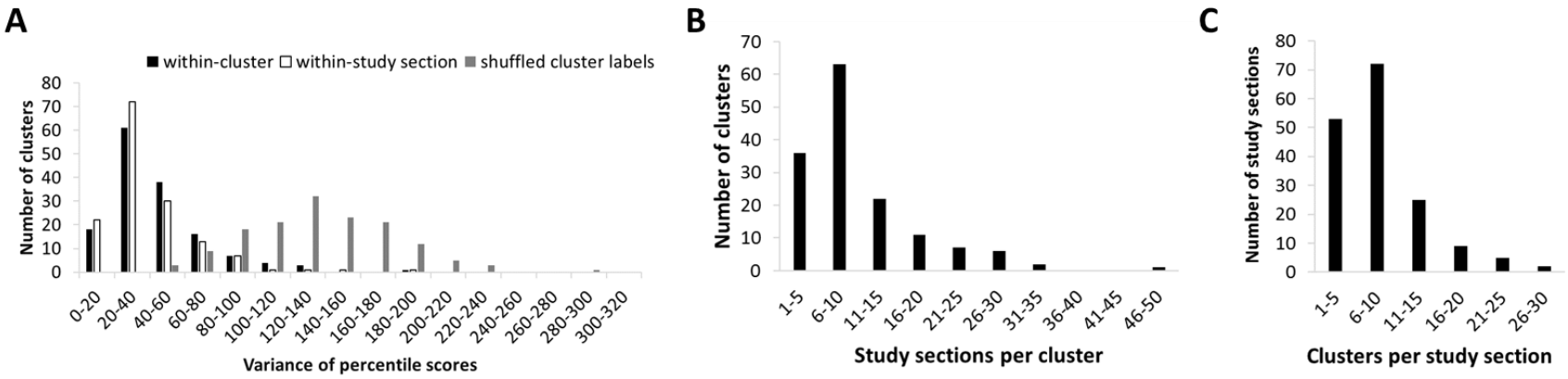
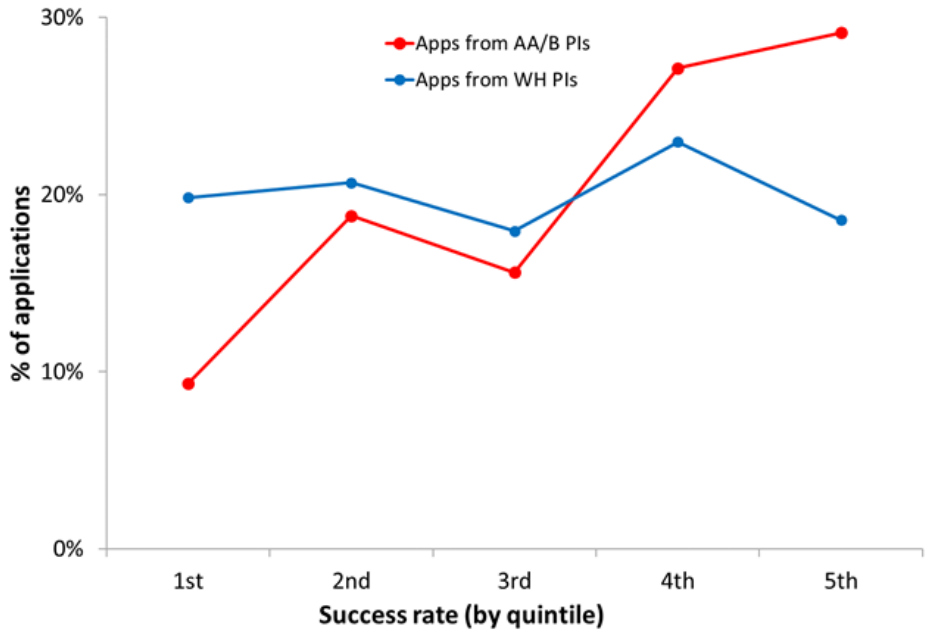


Fig. S5. Distribution of topics across study sections. (**A**) Variance of percentile scores by study section for a given topic cluster (weighted by the number of study sections represented in each cluster; "within-cluster") and by topic cluster for a given study section ("within-study section") have similar distributions ($p>0.05$). A shuffled dataset across all study sections was created by randomizing cluster labels, and differs significantly from within-cluster and within-study section variances ($p<0.0001$). (**B**) Number of study sections per topic cluster. (**C**) Number of topic clusters per study section.



	Award Rate (by quintile)	
	AA/B	**WH**
1st	13.4%	24.2%
2nd	12.6%	18.9%
3rd	13.6%	16.8%
4th	10.6%	15.3%
5th	7.0%	13.1%
Overall	10.7%	17.7%

Fig. S6. Comparison of topic choice variation by race. Topic clusters were separated into quintiles by their award rates (quintile 1 is the highest-success quintile). Award rate by quintile for applications from AA/B vs. WH applicants are displayed in the table on the right.

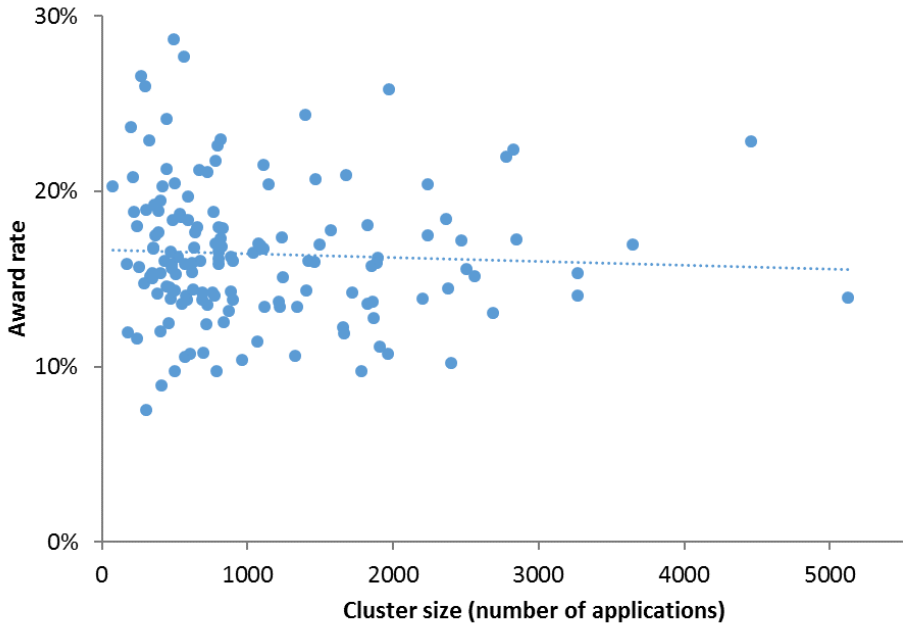
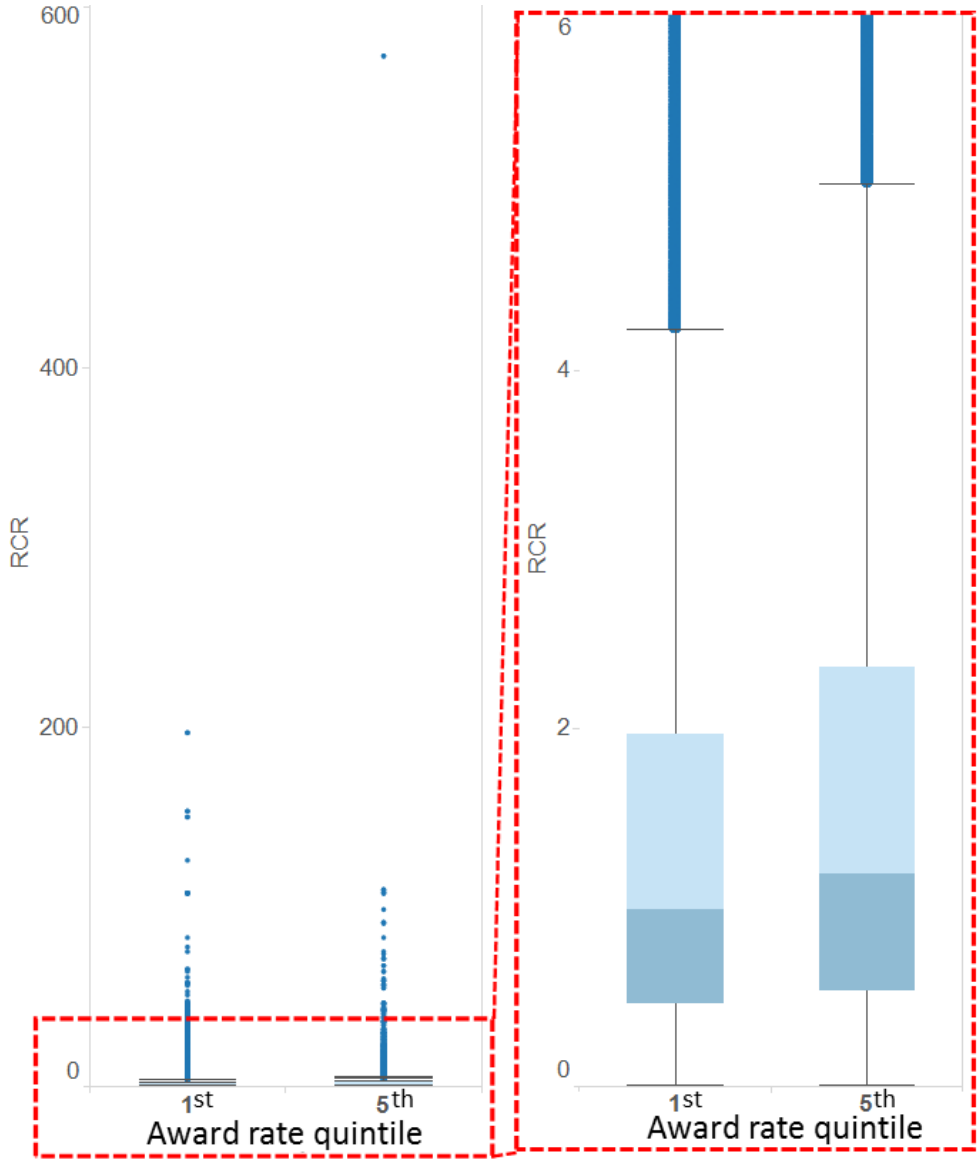


Fig. S7. Award rates by topic cluster size. Award rates are not correlated with topic cluster size, as measured by number of applications in each cluster (slope = -2×10^{-6}, $r^2 = 0.0023$).



		Award rate quintile 1st	Award rate quintile 5th
Number of Publications		116,358	60,742
Citations per year	Maximum	3,232	1,488
	Mean	17.94	16.67
	S.E.M.	0.12	0.15
	Median	9	8
Relative Citation Ratio (RCR)	Minimum	0	0
	Lower quartile	0.46	0.53
	Median	0.99	1.19
	Upper quartile	1.97	2.34
	Maximum	197.22	573.29
	Mean RCR	1.70	1.99
	S.E.M.	0.01	0.02
	Weighted RCR	197,377	120,802

Fig. S8. Scientific influence for higher and lower success topics. The higher the Relative Citation Ratio (see Methods), the more highly-cited papers are after accounting for area of research. Higher-success topics are the 20% of topic clusters with the highest award rate (1st quintile); lower-success topics are those in the 20% of topic clusters with the lowest award rate (5th quintile).

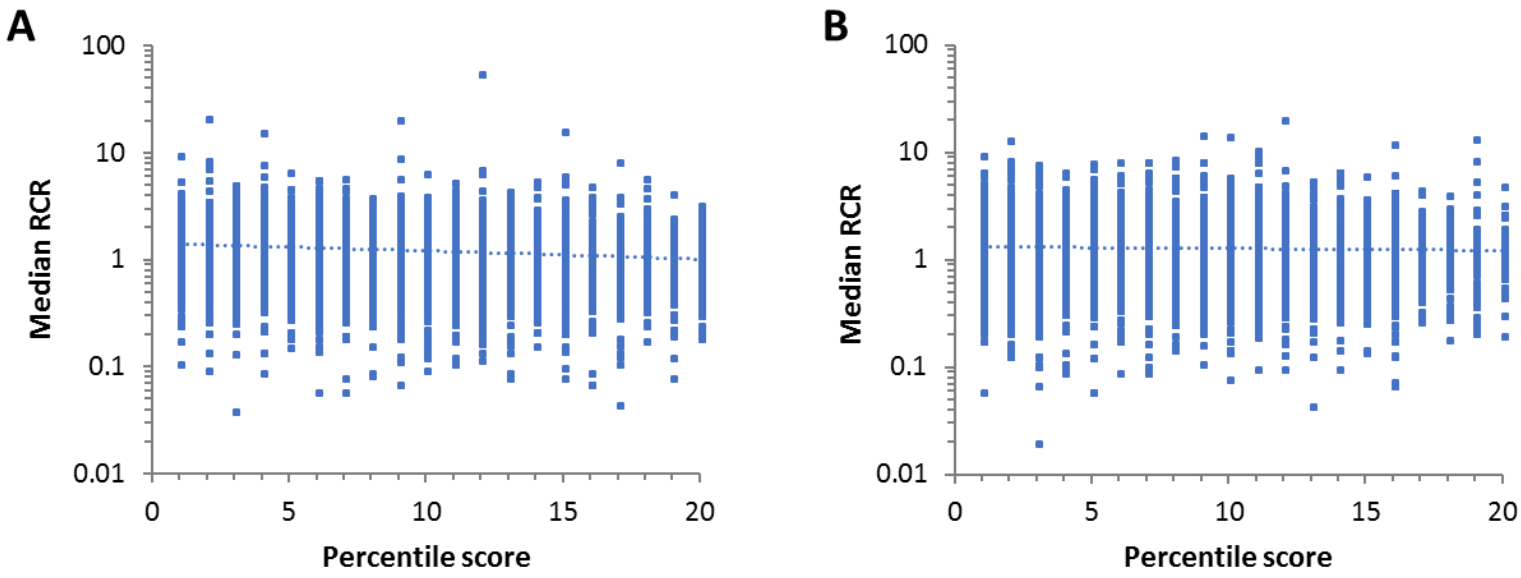


Fig. S9. Comparison of scientific influence for publications linked to awards from higher and lower success topics by percentile score. Percentile scores of awards are not correlated with influence (median RCR) of publications linked to those awards in either higher- (**A**) orlower-success (**B**) topic areas ($r^2 < 0.01$). Higher-success topics are the 20% of topic clusters with the highest award rate; lower-success topics are those in the 20% of topic clusters with the lowest award rate.

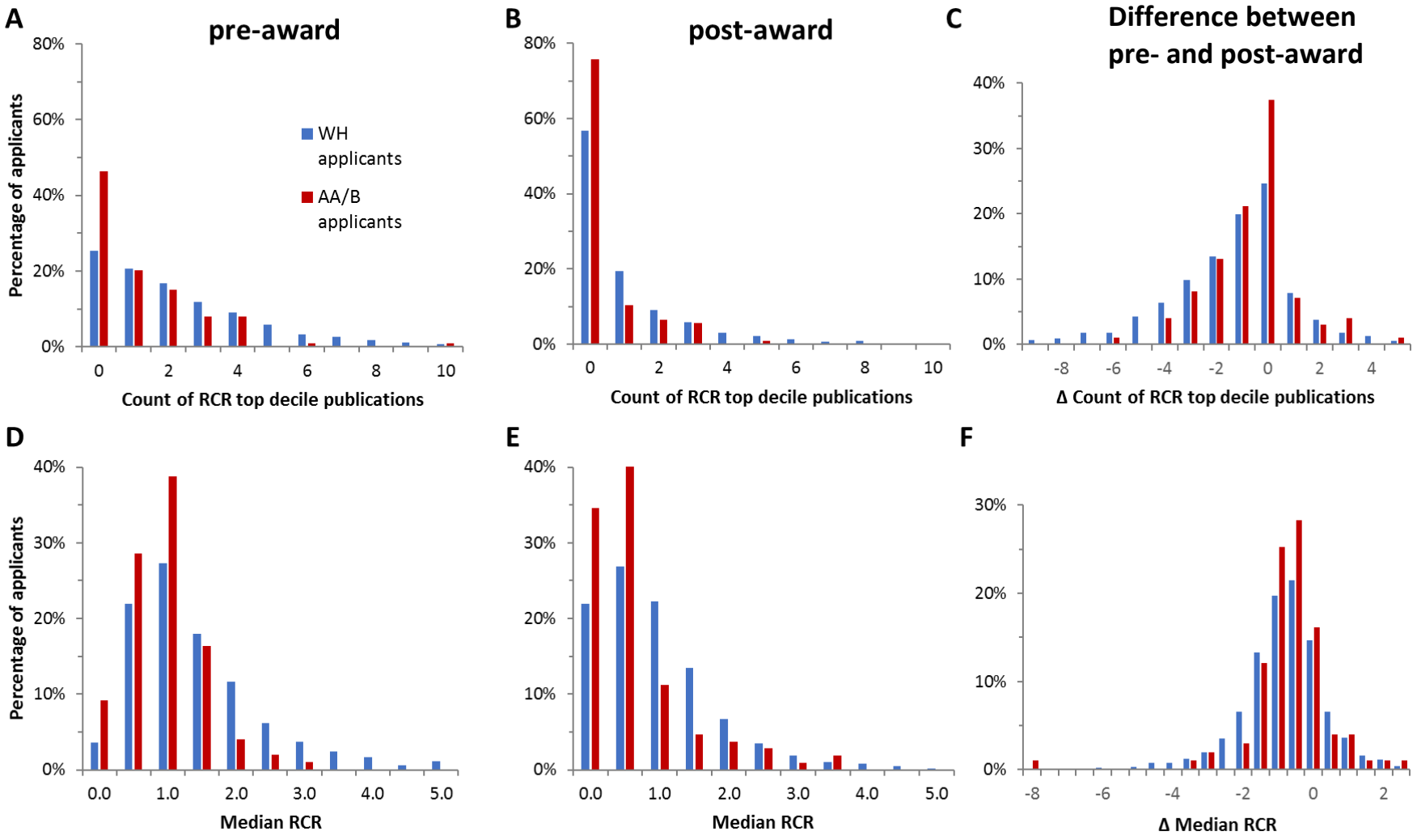

A
pre-award
Percentage of applicants
80%
60%
40%
20%
0%
WH applicants
AA/B applicants
0
2
4
6
8
10
Count of RCR top decile publications
B
post-award
80%
60%
40%
20%
0%
0
2
4
6
8
10
Count of RCR top decile publications
C
Difference between pre- and post-award
40%
30%
20%
10%
0%
-8
-6
-4
-2
0
2
4
Δ Count of RCR top decile publications
D
Percentage of applicants
40%
30%
20%
10%
0%
0.0
1.0
2.0
3.0
4.0
5.0
Median RCR
E
40%
30%
20%
10%
0%
0.0
1.0
2.0
3.0
4.0
5.0
Median RCR
F
30%
20%
10%
0%
-8
-6
-4
-2
0
2
Δ Median RCR

Fig. S10. Scientific influence of publications by new investigators before and after receiving their first award. Publications from new investigator's pre-award biosketches (**A**, **D**)and publications linked to their NIH award after receiving their first award (**B**, **E**) were evaluated for count of publications with an RCR in the top decile of NIH publications (RCR>3.81; A, B) and median RCR (D, E).The delta between pre- and post-award values for either count of publications with an RCR in the top decile, or median RCR, are shown in (**C**) and (**F**), respectively.